

COSMO® Cube

Hardware Plan Preview v1.0

LOT Systems / COSMO®
Prepared 2026-07-17
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STATUS: PLAN — pre-procurement. No parts ordered, no PCB fabricated, no stainless steel machined, no funds committed.
This document is a buildable reference, not a record of a completed build.

The Concept

COSMO® Cube is a small physical companion object for LOT — a "pager" for proactive AI notifications from lot-systems.com. It sits on a desk. Most of the time its screen is dark. When the LOT Quantum Intent Engine decides a user needs a nudge — "Coffee time!", "Breathe.", "You've been at this 90 minutes." — the Cube's screen lights up with that one line. A single button sends one signal back: **Copy** — "seen, acknowledged" — which lands as a Log entry the user can see in their own Log tab on lot-systems.com.

It is deliberately not a general-purpose IoT gadget. One notification channel in, one button out, a camera and a weather sensor for local context, and nothing else. The restraint is the design.

Physical Form

Two-part stainless steel body. The top shell is a flat, mirror-polished silver square — 40mm × 40mm × 5mm — that reads as an inert object on the desk when idle, not a lit gadget. The bottom shell is the instrument face: a 1.28" round display, a small camera, and a single tactile button.

#	Requirement	Design decision
1	PCB Way	Single vendor for PCB fab + assembly + CNC stainless steel enclosure
2	Pager-like AI notification	LOT server pushes a short string to the device over Wi-Fi; screen wakes, shows text, dims
3	Two-part stainless steel body	Top shell (polished plate) + bottom shell (instrument face), joined with hex standoffs + gasket
4	Flat silver square 4x4cm x 5mm	The top shell itself — #8 mirror polish, doubles as a passive desk mirror when dark
5	Camera	OV2640 on the MCU module — ambient context signal, off by default
6	LOT API connector	Firmware talks to a new /api/device/* surface on lot-systems.com
7	Result in PDF manuals	Assembly manual + user manual, generated from Markdown source
8	Compress info per session	Handled by the repo's existing Self-Assembly ledger/lexicon/doctrine protocol
9	Firmware documents	Separate firmware specification document
10	Software to connect with firmware	Separate software/API integration document
11	Separate documents	Plan / BOM / firmware / software / manufacturing / manual kept as distinct files
12	Charger	Qi wireless coil + USB-C wired fallback for flashing/debug
13	100-unit run	Costed as a distinct pilot-run tier in the BOM and manufacturing plan
14	Weather sensor	BME680 — temperature / humidity / pressure / VOC air quality
15	AI-grade off-the-shelf sensors	BME680 + OV2640 — standard, well-documented, widely-stocked parts
16	"Copy" button to Log tab	Button press posts to the device API, writes a Log row, appears in the existing Log tab feed
17	One side polished stainless steel	Top shell, #8 mirror finish
18	Other side: camera, screen, button	Bottom shell — the instrument face

#	Requirement	Design decision
19	Wireless charging	Qi receiver module + LiPo pouch cell

Roadmap

Phase 0 — Plan (this session)

Concept, physical form spec, bill of materials with real vendor links, firmware architecture spec, software/API integration spec, manufacturing plan, manual outline. Requires S-2 sign-off before proceeding — ordering parts and machining metal are irreversible, money-spending actions.

Phase 1 — Proof of unit (1 unit, ~2–3 weeks)

Order BOM parts for one prototype at dev-board stage. Breadboard build using the camera/MCU module, round display, environmental sensor, Qi receiver, and button. Flash firmware v0.1 and validate the LOT API surface end-to-end.

Phase 2 — First custom PCB (1–5 units, ~3–4 weeks)

Design one custom PCB integrating the module, display connector, sensor, Qi receiver, and button. Order via PCBWay prototype PCB + SMT assembly. Validate fit with a temporary 3D-printed enclosure before committing to metal.

Phase 3 — Enclosure and finish (parallel to Phase 2, ~2 weeks)

Finalize the two-shell stainless steel CAD. Order one unpolished test set, then one mirror-polished set, to confirm fit and finish before the full run.

Phase 4 — Pilot run (100 units)

Lock BOM, PCB revision, and enclosure CAD. Single PCBWay order for 100x PCB assembly and 100x CNC stainless steel shell pairs. Manual bench assembly in batches of roughly ten, with a five-point function test per unit.

Phase 5 — Field

Ship to first users. Firmware OTA update path for post-ship fixes. Every firmware or software revision runs through the existing Self-Assembly session-report protocol, same as every other LOT feature.

Estimated total timeline, sign-off to shippable units: 11–15 weeks, assuming no design revisions between phases. Full bill of materials, firmware spec, software integration spec, and manufacturing detail are maintained as separate Markdown documents in docs/hardware/ in the LOT-Computer repository.